- Algorithm: A process for solving a problem that is defined by a series of steps.

Data Structure: A special way to store in a computer memory and to structure data in order to help it be stored and accessed quickly with ease. A finite number of well-defined instructions that will solve a particular problem or perform a computation.Algorithms – The logic used to act on the data, Data structures – The structure used to store the data. Combined, they are a fundamental part of software design and computer science, directly affecting the performance and the use of resources of the program.

- Linear data structure is a data structure with data elements placed in adjacent linear order that are linked sequentially or linearly to each of the two adjacent data elements. Overall an element is accessed sequentially 1 pass. Memory can be implemented contiguously (e.g., like an array) or not (e.g., like a linked list). Such as Array, Linked List, Stack, and Queues.

- One-Dimensional Array is a linear array of elements which are all stored in contiguous (sequential) memory locations of the same data type (or class); a fixed-size array. Zero-based addressing. Direct access to any element is possible with constant time as memory locations are known directly. Insertion and deletion operations, on the other hand, involve shifting of elements and hence take time.

- Big O Notation is crucial in computer science as it provides a descriptive function, or upper bound on an algorithm's worst case time and space complexity as n increases. Big O measures how execution time or memory usage scales relative to input size, ignoring hardware variations and constant factors.

Constant: Execution time stays the same regardless of input size (e.g., array index lookup).
Logarithmic: Execution time grows logarithmically by halving the search space (e.g., binary search).
Linear: Execution time grows proportionally with input size (e.g., linear search).
Linearithmic: Standard complexity for efficient sorting algorithms (e.g., Merge Sort, Quick
Quadratic: Execution time grows quadratically with nested loops (e.g., Bubble Sort).

- Basic Searching Algorithms

Searching algorithms are the system of procedures followed in locating a particular element of a set of elements called a target value.

Linear Search: Definition is a sequential search whereby each element in the list is examined in turn from the start to the end of the list until the data value is located and found, or until the end of the list.Overview: Works on both sorted and unsorted data structures. Its worst case time complexity is O(n) (if the target element is at the end or not in the array) and best case time complexity is O(1) (if the target element is the first one).
Binary Search is an efficient searching method on the principle of divide and conquer approach which divides the range at half level to find the target element repeatedly. Must have sorted data structure before searching. It does this by comparing the target to the middle element: If the two

are the same, the search is over, if they are smaller or larger, it takes away half of the array where the target will never be found. Its time complexity in the best case is O(1) and worst/average case time complexity is O(log n).

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